

PHASE 1 NREM SLEEP

There are four stages of NREM:

Stage 1: This is the lightest form of sleep and usually lasts for only a few minutes

Stage 2: Brain waves start to slow down, and there are short bursts of neural activity, caused by the brain organising memories and thoughts from the day. Takes up most of the sleeping hours and the body goes through multiple rounds of it

Stage 3: Deepest sleep stage, when the body works to repair injuries, build immunity and regulate cognitive abilities and memory. The body tries to get as much of this as possible. Stage 3 NREM and REM sleep take up 25% each of sleep time

Return to Stage 2: There is a momentary return to stage two NREM sleep before the body enters REM sleep

PHASE 2 REM SLEEP

This is when the brain dreams the most and brain activity is similar to when the body is awake. After this, the body returns to Stage one NREM and a new sleep cycle begins.

SLEEP DISORDERS

INSOMNIA

A condition where the body has trouble falling and/or staying asleep. As a result the person's overall sleep quality is poor. Related to poor sleeping habits, mental disorders, side effects of medication, overuse of blue light screens–such as mobile phones–or chronic illness

HYPERSOMNIA

The inability to stay awake during the day despite adequate amount of nighttime sleep. Causes include jet lag, alcohol use, extreme exertion, pregnancy, or narcotics use

PARASOMNIA

A condition where the body exhibits abnormal movements, behaviors, emotions, perceptions and dreams while falling asleep, sleeping, and/or between sleep stages. A person suffering from this condition may appear awake, when they are in fact, asleep. Signs include sleep walking, nightmares and talking in sleep

OBSTRUCTIVE SLEEP APNEA

Medical condition where the body has difficulty breathing in sleep. Manifests itself as snoring. Most common cause is obesity, as fat deposits tend to block the airways

SLEEP PARALYSIS

Afflicted person is unable to move their body parts even after waking up, because muscle atonia (when the brain signals the muscles to relax during REM sleep) lingers even after waking up

RESTLESS LEG SYNDROME (RLS)

Occurs while resting or sleeping when the person feels a constant urge to move their legs. Long-term health conditions and low levels of iron are possible triggers; worsens with age

NARCOLEPSY

Afflicted person may feel rested after waking up, but then get sudden attacks of sleepiness through the day. This is because the brain is unable to regulate the sleep-wake cycle. Exact cause not known, but genetics is a suspect

REM SLEEP BEHAVIOUR DISORDER

Characterised by the acting out of dreams by thrashing about, hand gesturing, punching, kicking or screaming during REM sleep phase. Trigger unknown

PERIODIC LIMB MOVEMENT

Results in the afflicted person repeatedly moving their leg and/or arm during all sleep cycles. Usually happens to people with restless leg syndrome, but not everyone with RLS will exhibit this sleep disorder

CIRCADIAN RHYTHM SLEEP DISORDER

The body's internal clock, called a circadian clock, is usually set on a 24-hour cycle, aligned to daylight. In some people this is out of sync with the environment This impacts their timing of sleep. It is a rare disorder and usually linked to genetics or other disorders